



DIOGO HENRIQUE SILVESTRE MATOS CEBOLA

PRIVACY-ENHANCED ONTOLOGIES

A STUDY ON SECURE PROTOCOLS FOR EXPRESSIVE SPARQL
QUERIES OVER ENCRYPTED ONTOLOGIES

MASTER IN COMPUTER SCIENCE

NOVA University Lisbon

Draft: February 14, 2022



PRIVACY-ENHANCED ONTOLOGIES

A STUDY ON SECURE PROTOCOLS FOR EXPRESSIVE SPARQL QUERIES OVER
ENCRYPTED ONTOLOGIES

DIOGO HENRIQUE SILVESTRE MATOS CEBOLA

Advisers: Matthias Knorr
Auxiliar Professor, NOVA University Lisbon
João Leitão
Auxiliar Professor, NOVA University Lisbon
Henrique Domingos
Associate Professor, NOVA University Lisbon

CONTENTS

List of Figures	iv
List of Listings	v
Acronyms	vi
1 Introduction	1
1.1 Context	1
1.2 Problem	3
1.3 Objectives	4
1.4 Contributions	4
1.5 Document Organization	4
2 Related Work	6
2.1 Semantic Web: A layered architecture	6
2.1.1 RDF: Resource Description Framework	8
2.1.1.1 Overview	8
2.1.1.2 Syntax	8
2.1.2 RDFS: RDF Schema	9
2.1.2.1 Overview	9
2.1.2.2 RDFS primitives	9
2.1.3 OWL2: Web Ontology Language	10
2.1.3.1 Overview	10
2.1.3.2 OWL2 DL	11
2.1.3.3 Owl Language	11
2.1.4 SPARQL	13
2.1.5 Tools & Development Software for Ontologies	13
2.2 Secure Computations	13
2.2.1 Homomorphic Encryption	13
2.2.2 Searchable Encryption	13

2.2.3	Symmetric Searchable Encryption	13
2.2.3.1	Structured Searchable Encryption	13
2.2.3.2	Dynamic Symmetric Searchable Encryption	13
2.3	Summary	13
3	Approach to Elaboration	14
3.1	System Requirements	14
3.2	Privacy Guaranties	14
3.3	Architecture & System Model	14
3.3.1	Implementation Guidelines	14
3.3.2	Validation	14
4	Work Plan	15
	Bibliography	16

LIST OF FIGURES

2.1 The <i>Semantic Web</i> stack [47].	7
---	---

LIST OF LISTINGS

2.1	Simple RDF Example, written in Terse RDF Triple Language (TURTLE)	
	syntax.	9

ACRONYMS

DSSE	Dynamic Symmetric Searchable Encryption 3 , 4
FHE	Fully Homomorphic Encryption 3 , 13
GO	Gene Ontology 1
HE	Homomorphic Encryption 2 , 3 , 6 , 13
HTML	HyperText Markup Language 9
LFHE	Leveled-Fully Homomorphic Encryption 3 , 13
NCI	National Cancer Institute 1
ORAM	Oblivious RAM 3
OWL2	Web Ontology Language 7 , 8 , 10 , 11
PEKS	Public Key Encryption with Keyword Searchable 3 , 13
PHE	Partially Homomorphic Encryption 3 , 13
RDF	Resource Description Language vi , 3 , 7 , 8 , 9 , 10 , 11
RDFS	Resource Description Framework Schema 7 , 8 , 9 , 10
SE	Searchable Encryption 2 , 3 , 6 , 13
SHE	Somewhat Homomorphic Encryption 3 , 13
SLA	Service Level Agreement 2
SPARQL	SPARQL Protocol and RDF Query Language 3 , 4 , 7 , 8
SSE	Symmetric Searchable Encryption 3 , 4 , 13
STE	Structured Symmetric Encryption 3 , 4

SWRL	Semantic Web Rule Language 7
TURTLE	Terse RDF Triple Language v , 8 , 9
URI	Uniform Resource Identifiers 6 , 8
W3C	World Wide Web Consortium 6
WWW	World Wide Web 1
XML	Extensible Markup Language 7 , 9 , 11

INTRODUCTION

1.1 Context

In 2001, the term *Semantic Web* [4] was first used to define what would be the evolution of the [World Wide Web \(WWW\)](#). The traditional *Web* solved the challenge of distributing information at scale and globally, however it was designed for human use. Documents are easily accessible and readable by humans, but for automated agents little more than document retrieval is possible. Contrary to humans, machines cannot reason over the information they access, if they do not know the *semantics of the data*.

The *Semantic Web* addresses this issue, by redesigning the representation of information, in a way that not only preserves meaning and relationships but is also machine-readable. A concept is mapped to a unique identifier, and meaning is encoded in sets of triples, each triple consisting of a subject, predicate and object (similar to an elementary sentence). *Ontologies* provide a shared understanding of a domain by formalizing the specification of relevant concepts, and their relations, in such domain.

This novel method of knowledge representation differs from the previous, centralized way, and greatly simplifies the process of interconnecting different knowledge bases. Centralized representations tend to be more expensive to develop, usually needing to be built from scratch [20]. Furthermore, for sharing, both parties need to agree on the semantics. In contrast, using ontologies, information is semantically linked by default. In the *Semantic Web*, machines can truly communicate, share and reason about distinct knowledge bases.

Knowledge representation is essential in the industry [21] and ontologies help remove ambiguity in terminology. Nowadays, they are being developed and applied in a variety of domains [46] ranging from the health sector¹, to linguistics² and even being used by companies like NASA³ [3].

¹The NCI Thesaurus [27], developed by the [National Cancer Institute \(NCI\)](#) or the [Gene Ontology \(GO\)](#) [13], the world's largest source of information on the functions of genes, are some existing applications.

²The WordNet [17] is a lexical database for English where related words and concepts are semantically linked.

³Observing the visual representation provided by the Linked Open Data Cloud [16] is a good exercise to perceive the dimension of adoption of this new paradigm.

To develop and maintain ontologies, advanced semantic editors are the standard tool for professionals, with Protégé [19] being a very popular example. These provide numerous functionalities, usually based on plugins (e.g. automatic verification of errors and consistency in the design; debugging tools; query answering; graphic visualization of data).

With ontologies gradually growing in dimension, the industry is currently evolving towards the use of collaborative solutions [32, 31] and outsourcing heavy computations, relying on cloud infrastructures or other shared computing platforms. These approaches, albeit inevitable, present new security concerns and engineering challenges.

As of 2021, cloud adoption has risen more than what was previously forecasted, with 50 percent of all enterprise workloads hosted in *public clouds* and predictions of 7 percentage point gain in 2022 [26]. Research on secure cloud computing [7] studies such adoption but also discusses security concerns. Adoption can be explained with the cost efficiency⁴, high scalability, flexibility and availability that cloud provides. Additionally, the pay-per-use model is an easy entry point to the ecosystem.

On the contrary, while delegation of responsibilities is beneficial for costs, it is prejudicial to control. Using cloud services implies trusting its owner. A [Service Level Agreement \(SLA\)](#) defines, in accordance to metrics, the service levels of the system that the vendor should provide (e.g. percentage of availability), while also defining penalties in case of not achieving what was agreed-on. However, the risk of loss of privacy and data breaches still exists. A problem that is especially concerning when sensitive data, such as health records, is being stored in such services. This is much more relevant in *public clouds* where, compared to *private clouds*, trust is distributed between various actors and not on a single organization [7].

In cloud storage services, administrators, or hacker with root rights, might have access to the data [5]. Additionally, these cloud services might outsource computations to third parties whom we do not trust. In this setting, we usually consider the *honest-but-curious*, or *passive*, adversary model. This adversary will follow the protocol properly but keep a record of all its intermediate computations [11]. Furthermore, it cannot do any other action except attempting to learn private information while observing views of the protocol execution [9].

A first approach to secure data in the cloud would be to use *Encryption at Rest*, which naively encrypts all data stored in the untrusted servers. If a user wants to do a write or search operation over the data, they will have to download all the data, and decrypt it. This incurs in high network requirements and is computationally expensive on the client-side, hindering all the benefits of using a cloud service to store the data. Nonetheless, this is the mechanism usually provided by database services [23, 22, 18].

Recently, active research is being done in the fields of [Searchable Encryption \(SE\)](#) and [Homomorphic Encryption \(HE\)](#).

⁴Cloud services reduce implementation and maintenance, power, cooling, floor space, storage and operational costs.

Homomorphic Encryption is a cryptographic method based on the concept of algebra homomorphism, where both decryption and encryption can be seen as an homomorphism of the relations between the plaintext and the ciphertext [1, 10]. **Homomorphic Encryption**, based on the type and number of operations allowed to be done on the encrypted data, can be classified as: **Fully Homomorphic Encryption (FHE)** supporting any number of operations with no restriction on the number of times; **Somewhat Homomorphic Encryption (SHE)** permitting some types of operations, a limited number of times; **Partially Homomorphic Encryption (PHE)** allowing one type of operation with no restriction on the number of times; and **Leveled-Fully Homomorphic Encryption (LFHE)** [6] supporting any number of operations, but with a pre-determined expected depth- L .

Searchable Encryption uses classical encryption schemes to perform encrypted search queries, sequentially over the ciphertext [28], or through the generation of searchable *encrypted indexes* accessible with *tokens*, generated via *trapdoor* functions. **Searchable Encryption** can be divided in **Symmetric Searchable Encryption (SSE)**, when it is based on symmetric key cryptography, and **Public Key Encryption with Keyword Searchable (PEKS)**, when asymmetric key algorithms are used. These techniques may leak some information about the *indexes*, *search pattern*, and *access pattern* during execution. Most approaches leak, at least, the *search* and *access patterns* [5].

Besides **HE** and **SE**, **Oblivious RAM (ORAM)** [12] could theoretically be used as an alternative. However, **ORAM** and **HE** are not efficient enough, when used exclusively, to be feasible yet [5] in real world applications. Also, **SSE** tends to be more practical than **PEKS**, due to the efficiency comparison between asymmetric and symmetric cryptographic schemes [24].

1.2 Problem

Ontologies usually only contain public, and non-sensitive data, but sometimes they are used alongside electronic health records and clinical decision systems. Furthermore, they can hold private value, due to the investment of time and money in their development.

Taking into considerations the efficiency and security trade-offs from the various alternative techniques, the **SSE** approach seems to be the most practical solution for protecting ontologies in untrusted servers. However, most **SSE** protocols only consider *unstructured* collections of documents with no semantic relationships. More recently, Chase and Kamara, on their work on **Structured Symmetric Encryption (STE)**, generalized index-based **SSE** to support queries on *structured* data [8].

Ontologies can be represented as graphs and **STE** supports various types of query operations in graph-structured data. Nevertheless, **STE** with direct support for the query expressiveness required for searches over ontologies has not yet been studied, more specifically for languages like **SPARQL Protocol and RDF Query Language (SPARQL)** [25, 30].

Furthermore, write operations that are supported in **Dynamic Symmetric Searchable Encryption (DSSE)** [15], have not yet been studied in the context of ontologies.

Additionally, information leakage can be a problem while trying to guaranty privacy of the topology of an ontology.

To conclude, we will focus on answering the following questions:

1. *How to support expressive queries and other operations in privacy-enhanced encrypted ontologies, using techniques such as [Structured Symmetric Encryption](#) and [Dynamic Symmetric Searchable Encryption](#)?*
2. *Besides preserving data privacy, how can we minimize information leakage and avoid possible exposure of the topology of the encrypted ontologies?*

1.3 Objectives

The main objective of this thesis will be to provide answers to the two questions defined in section 1.2 through a problem driven approach and guide future work in this field of research. We will first study a static [STE](#) approach focusing on support for [SPARQL](#) queries in a multi-user setting, later expanding to a dynamic setting. Finally, we will study how to minimize information leakage about the topology of the encrypted data.

1.4 Contributions

Our contributions will be the proposal of a protocol that extends upon [STE](#) and supports expressive and secure [SPARQL](#) queries over privacy-enhanced encrypted ontologies. We now present a summary of the contributions:

SSE M/M Protocol for Privacy-Enhanced Ontologies A novel multi-user protocol, based on [STE](#), that supports [SPARQL](#) queries over encrypted ontologies, stored in un-trusted servers.

P1 - Static Setting A protocol implementation for a static setting, supporting only search operations, based on [STE](#).

P2 - Dynamic Setting A protocol implementation for a dynamic setting, supporting search, write and delete operations, based on [STE](#) and [SSE](#).

All the prototypes developed will be available as open-source code in a public repository, alongside the benchmark results.

1.5 Document Organization

The remaining of the document will be organized as follows:

Chapter 2 Analyses related work on *Semantic Web, Ontologies and Searchable Encryption*.

Chapter 3 Presents requirements and informally defines the protocols and system models. Furthermore, in this chapter we describe our implementation and testing approaches.

Chapter 4 Describes the work plan of the elaboration phase tasks.

RELATED WORK

On this chapter we will discuss the architecture and design of the *Semantic Web* as well as analyzing work already done in the fields of *Secure Computations*. The remaining of the chapter is organized as follows: on section 2.1 we discuss the *Semantic Web* stack, languages and tools; section 2.2 describes work on *Secure Computation*, with particular attention regarding HE and SE; finally, on section 2.3 we summarize and discuss which work is relevant.

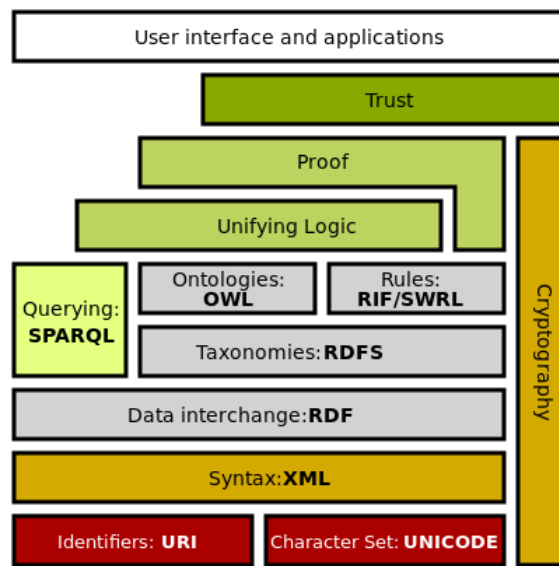
2.1 Semantic Web: A layered architecture

In section 1.1, we have contextualized the main philosophy and design principles of the *Semantic Web*:

1. Usage of *labeled graphs* as the data model where concepts are represented as triples consisting of a subject, predicate and object (with nodes being objects and relations between such objects modelled as the edges);
2. All data items are identified by [Uniform Resource Identifiers \(URI\)](#);
3. Usage of ontologies to model semantics within a domain.

Additionally, the *Semantic Web* can also be seen as a stack of various layers [2] (figure 2.1). Most of these layers are standardized due to the collective work of the [World Wide Web Consortium \(W3C\)](#).

Briefly describing this layered architecture, at the root level, we have the [URIs](#) and a character set providing the baseline of communication, with distinguishable identification of resources and an agreed encoding.

Figure 2.1: The *Semantic Web* stack [47].

At the syntax level we find the [Extensible Markup Language \(XML\)](#) [33]. XML supports the definition of unlimited *tags*, that can be placed in the text of a document. These *tags* facilitate the creation of hierarchy and give *structure* to the information.

From structured data we then define the basic data model using [RDF](#) [40]. With a XML like syntax, we can write statements about resources. Thus, embedding *semantics* within the information, as *metadata*.

The [Resource Description Framework Schema \(RDFS\)](#) [44] is a vocabulary description language to organize [RDF](#) resources into hierarchies, based on primitives such as *classes* and *properties*. It can be seen as a rudimentary *ontology language* [2].

Ontologies are defined with the [Web Ontology Language \(OWL2\)](#) [35]. Like [RDFS](#), it is also a vocabulary description language, but provides more expressiveness (e.g. relations between classes, *cardinality*, *equality*, richer properties with characteristics).

[RDF](#) triples can be interpreted as logical binary predicates, making reasoning possible over [RDF](#)-based data. This allows us to infer and validate ontologies, and also provide explanations on operations' results. The rules layer is a reference to this specific part of the architecture stack. Many rule languages, like the [Semantic Web Rule Language \(SWRL\)](#) [14], facilitate interaction based on *descriptive-logic*.

To query and retrieve any data based on [RDF](#) we use [SPARQL](#).

The Proof and Unifying Logic layers include the notion of interconnectivity of terminologies between *knowledge bases*, with explainable proofs to operations.

Only with secure cryptographic schemes can we reach the trust layer. The security layer is still not standardized.

The top layer, references all the applications and interfaces that use this new paradigm of data representation.

In the following of this section we will describe, in more detail, some languages and

tools. We will focus on [RDF](#) in section 2.1.1; on [RDFS](#), in section 2.1.2; on [OWL2](#), in section 2.1.3; on [SPARQL](#), in section 2.1.4; on common tools and software, in section 2.1.5.

2.1.1 RDF: Resource Description Framework

2.1.1.1 Overview

[RDF](#) [40] is a flexible *domain independent* data model. The [RDF](#) basic primitives are *resources*, *Properties* and *literals*. We can construct *statements* about resources by organizing these basic primitives in sets of *triples*. A triple is composed by a *subject*, a *predicate* and an *object*. In a statement (triple) about a subject (always a resource) we define the value (object) of a property (predicate), which can be another resource or a literal.

Literals are atomic values (e.g. strings, numbers). Resources and properties, respectively, model the "things" and relationships between such "things". Both are identified using [URIs](#). Resources without a [URI](#) are called *blank nodes* (or *bnodes*).

To make a statement *B* about a statement *A* we can use a technique called *reification*. Reification consists in: firstly, defining a statement *B* where the object *objA* represents the statement *A*; secondly, creating the properties *subject*, *predicate* and *object* mapped from *objA* to the original subject, predicate (now a resource), and original object of statement *A* [2].

2.1.1.2 Syntax

Graphical [RDF](#) being a data model, can be written using different syntaxes. Graphically we can represent statements as a labeled and directed graph. The *nodes* are labelled with the resources [URIs](#), a literal value or left blank, whereas *arcs* are labelled with the properties [URIs](#). The arcs are directed from subject to object of a statement.

1) TODO: ADD graphical example

T¹

Text-Based There are multiple text-based syntaxes for [RDF](#). Some of the most commonly used are [TURTLE](#) [42] (listing 2.1), and its extension for named graphs *TriG* [41]. In [TURTLE](#), [URIs](#) are enclosed by angle brackets. The subject, property, and object are written in this order. Statements are finished by a period. Literals are surrounded in quotes with the type of data specified with a [URI](#). The data value and type are separated by ^^ (if no data type is mention, the value is inferred as being a string). We can also define [URI](#) abbreviations (using @prefix, followed by the prefix to substitute the [URI](#), a colon, the [URI](#) and a period) and simplify statements, for easier readability (e.g.using a semicolon to make more than one statement about the same subject, defining statements with nested blank nodes using square brackets). Comments start with #.


```

1 # FCT is located in Almada
2 <http://my-custom-domain/faculties.ttl#FCT> <http://my-custom-domain/ontology/location> <
   ↪ http://my-custom-domain/resource/Almada> .
3
4 # FCT has 8.000 students
5 <http://my-custom-domain/faculties.ttl#FCT> <http://my-custom-domain/faculties.ttl#
   ↪ asNumberOfStudents> "8000"^^<http://www.w3.org/2001/XMLSchema#integer> .

```

Listing 2.1: Simple RDF Example, written in **TURTLE** syntax.

XML/RDF [43] is a standardized **XML**-based syntax and *RDFa* [45] permits **RDF** to be embedded in the **HyperText Markup Language (HTML)**¹.

2.1.2 RDFS: RDF Schema

2.1.2.1 Overview

RDF is a flexible data-model because it is not domain-specified. However, for interconnectivity of different **RDF** knowledge bases, semantics for restricting meaning in a specific domain are needed.

RDFS [44] does that by providing vocabulary to describe groups of related resources and the relationships between these resources. This vocabulary is a set of **RDF** resources identified under the <http://www.w3.org/2000/01/rdf-schema#>.

2.1.2.2 RDFS primitives

Classes group resources together, the *instances* of the class. *Properties* are defined by specifying to which classes of resource they may apply, using the mechanism of *domain* and *range*. This allows for easy extension, by not having to redefine classes wherever a new property, that applies to an existing class, is needed.

To state that a resource is an instance of a class we use the *rdf:type* property.

The *rdfs:subClassOf* is used to define that all the instances of a class are instances its *superclasses*: if *A rdfs:subClassOf B*, then instances of *A* are also instances of *B*. The *rdfs:subPropertyOf* states that instances related by a property are also related by its *superproperties*: if *A rdfs:subPropertyOf B*, then instances related to *A* are also related to *B*. The properties *rdfs:subClassOf* and *rdfs:subPropertyOf* are *transitive*, and there are no restrictions on the number of superclasses or superproperties that, respectively, a class or a property can have.

The *rdfs:domain* property defines that any resource with a given property is an instance of the domain classes that the property applies to. The *rdfs:range* property specifies that the values of the property are instances its range classes.

In tables 2.1 and 2.2, we summarize the core **RDFS** classes and properties. In figure ..., we extend the **RDF** graphical example ... with **RDFS**.

¹We recommend the curious reader to read the W3C recommendations to better learn the syntaxes.

RDFS Core Classes	
Class	Description
rdfs:Resource	The class resource, everything.
rdfs:Class	The class of classes.
rdfs:Literal	The class of literal values, e.g. textual strings and integers.
rdf:Property	The class of RDF properties.
rdfs:Statement	The class of RDF statements.

Table 2.1: RDFS Core Classes and respective description (rdfs:comment).

Core RDFS Properties			
Property	Description	Domain	Range
Relationship Definition			
rdf:type	The subject is an instance of a class.	rdfs:Resource	rdfs:Class
rdfs:subClassOf	The subject is a subclass of a class.	rdfs:Class	rdfs:Class
rdfs:subPropertyOf	The subject is a subproperty of a property.	rdf:Property	rdf:Property
Property Restriction			
rdfs:domain	A domain of the subject property.	rdf:Property	rdfs:Class
rdfs:range	A range of the subject property.	rdf:Property	rdfs:Class
Reification			
rdf:subject	The subject of the subject RDF statement.	rdf:Statement	rdfs:Resource
rdf:predicate	The predicate of RDF statements.	rdf:Statement	rdfs:Resource
rdf:object	The object of RDF statements.	rdf:Statement	rdfs:Resource
Utility			
rdfs:seeAlso	Further information about the subject resource.	rdfs:Resource	rdfs:Resource
rdfs:isDefinedBy	The definition of the subject resource.	rdf:Resource	rdfs:Resource
rdfs:comment	A description of the subject resource.	rdf:Resource	rdfs:Literal
rdfs:label	A human-readable name for the subject.	rdf:Resource	rdfs:Literal

Table 2.2: RDFS Core Properties and respective description (rdfs:comment).

2.1.3 OWL2: Web Ontology Language

2.1.3.1 Overview

An *ontology* was defined by Studer et al. as "a formal, explicit specification of a shared conceptualisation" [29]. Meaning that an ontology should specify consensual and machine-readable knowledge by means of a formal definition of concepts, where concepts have explicit types and restrictions. Therefore, an *ontology language* should provide a well-defined syntax, formal semantics, convenient and sufficient expressiveness and efficient reasoning support.

RDFS can be considered a very limited *ontology language*. OWL2 [35] provides a number of additional features (e.g. cardinality constraints, class equivalence, intersection and disjunction). OWL2 does not simply extend RDFS because RDFS primitives are too expressive. This would make reasoning very inefficient and computationally expensive [2]. As such, OWL2 is divided in two sublanguages: *OWL2 Full* [37], a direct extension of RDFS with all OWL2 semantics; *OWL2 DL* [34], a language with some restrictions on the way OWL2, RDF and RDFS primitives may be used.

Contrary to *OWL2 Full*, *OWL2 DL*, being less expressive, retains *decidability*. The correspondence theorem of [34] states that: given an *OWL2 DL* ontology, inferences drawn using its *Direct Semantics* will still be valid if the ontology is mapped into an RDF graph and interpreted using the *RDF-Based Semantics*, the semantics of *OWL2 Full*.

OWL2 can be expressed with any valid *RDF* syntax, however there are various *OWL2* specific syntaxes: the Functional-Style Syntax [38], which is very compact and was the language used in the formal specification of the language; the Manchester Syntax [36], designed to be human-readable (it is very popular in user interfaces of semantic editors); and the OWL/XML Syntax [39] which permits compatibility with *XML* authoring tools.

2.1.3.2 OWL2 DL

2.1.3.3 Owl Language

Property Types

Object Properties

Datatype Properties

Annotation Properties

Top and Bottom Properties

Transitive Properties

Symmetric and Asymmetric Properties

Functional and Inverse-Functional Properties

Reflexive and Irreflexive Properties

Property Axioms

Domain and Range

Inverse Properties

Equivalent Properties

Disjoint Properties

Property Chains

Class Axioms

Subclass Relations

Class Equivalence

Punning

Enumerations

Disjoint Classes

Complement

Union and Disjoint Union

Intersection

Class Axioms on Properties

Universal Restrictions

Existential Restrictions

Necessary and Sufficient Conditions

Value Restrictions

Cardinality Restrictions

Data Range Restrictions and Datatypes

Self Restrictions

Individual Facts

Class and Property Assertions

Identity Assertions

Negative Assertions

2.1.4 SPARQL

2.1.5 Tools & Development Software for Ontologies

2.2 Secure Computations

2.2.1 Homomorphic Encryption

Homomorphic Encryption is a cryptographic method based on the concept of algebra homomorphism, where both decryption and encryption are seen as homomorphisms of the relations between the plaintext and the ciphertext [1, 10]. **HE** allows for computations done over encrypted data to be preserved after decryption of the ciphertext. An encryption scheme is homomorphic if such operations preserve additive or multiplicative properties. **HE** is classified in four main types: **FHE** supporting any number of operations with no restriction on the number of times; **SHE** permitting some types of operations, a limited number of times; **PHE** allowing one type of operation with no restriction on the number of times; and **LFHE** [6] supporting any number of operations, but with a pre-determined expected depth- L .

2.2.2 Searchable Encryption

Searchable Encryption uses classical encryption schemes to perform encrypted search queries. **Searchable Encryption** can be divided in **SSE**, when it is based on symmetric key cryptography, and **PEKS**, when public key algorithms are used. These techniques may leak some information about the *indexes*, *search pattern*, and *access pattern* during execution.

2.2.3 Symmetric Searchable Encryption

2.2.3.1 Structured Searchable Encryption

2.2.3.2 Dynamic Symmetric Searchable Encryption

2.3 Summary

APPROACH TO ELABORATION

3.1 System Requirements

3.2 Privacy Guaranties

3.3 Architecture & System Model

3.3.1 Implementation Guidelines

3.3.2 Validation

WORK PLAN

BIBLIOGRAPHY

- [1] A. Acar et al. “A survey on homomorphic encryption schemes: Theory and implementation”. In: vol. 51. 2018. DOI: [10.1145/3214303](https://doi.org/10.1145/3214303) (cit. on pp. 3, 13).
- [2] G. Antoniou et al. “A Semantic Web Primer (Third Edition)”. In: *The MIT Press* (2012), p. 287 (cit. on pp. 6–8, 10).
- [3] N. Ashish. “Semantic-Web Technology: Applications at NASA”. In: *Dagstuhl Seminar Proceedings*. 2005 (cit. on p. 1).
- [4] T. Berners-Lee, J. Hendler, and O. Lassila. *The semantic web*. Vol. 284. 2001. DOI: [10.1038/scientificamerican0501-34](https://doi.org/10.1038/scientificamerican0501-34) (cit. on p. 1).
- [5] C. Bösch et al. “A Survey of Provably Secure Searchable Encryption”. In: *ACM Comput. Surv.* 47.2 (Aug. 2014), pp. 1–51. ISSN: 0360-0300. DOI: [10.1145/2636328](https://doi.org/10.1145/2636328) (cit. on pp. 2, 3).
- [6] Z. Brakerski, C. Gentry, and V. Vaikuntanathan. “(Leveled) Fully Homomorphic Encryption without Bootstrapping”. In: *ACM Trans. Comput. Theory* 6.3 (July 2014), pp. 1–36. ISSN: 1942-3454. DOI: [10.1145/2633600](https://doi.org/10.1145/2633600) (cit. on pp. 3, 13).
- [7] M. Carroll, A. Van Der Merwe, and P. Kotzé. “Secure cloud computing: Benefits, risks and controls”. In: *2011 Information Security for South Africa - Proceedings of the ISSA 2011 Conference*. 2011. DOI: [10.1109/ISSA.2011.6027519](https://doi.org/10.1109/ISSA.2011.6027519) (cit. on p. 2).
- [8] M. Chase and S. Kamara. “Structured encryption and controlled disclosure”. In: *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. Vol. 6477 LNCS. 2010. DOI: [10.1007/978-3-642-17373-8_33](https://doi.org/10.1007/978-3-642-17373-8_33) (cit. on p. 3).
- [9] D. Evans, V. Kolesnikov, and M. Rosulek. “A Pragmatic Introduction to Secure Multi-Party Computation”. In: *Foundations and Trends® in Privacy and Security* 2.2-3 (2018), pp. 70–246. ISSN: 2474-1558. DOI: [10.1561/33000000019](https://doi.org/10.1561/33000000019) (cit. on p. 2).
- [10] C. Fontaine and F. Galand. “A survey of homomorphic encryption for nonspecialists”. In: *Eurasip Journal on Information Security* 2007 (2007). ISSN: 2510-523X. DOI: [10.1155/2007/13801](https://doi.org/10.1155/2007/13801) (cit. on pp. 3, 13).
- [11] O. Goldreich. *Foundations of Cryptography: Volume 2, Basic Applications*. en. Cambridge University Press, 2001. ISBN: 9780521830843 (cit. on p. 2).
- [12] O. Goldreich and R. Ostrovsky. “Software protection and simulation on oblivious RAMs”. In: *J. ACM* 43.3 (May 1996), pp. 431–473. ISSN: 0004-5411. DOI: [10.1145/233551.233553](https://doi.org/10.1145/233551.233553) (cit. on p. 3).

- [13] M. A. Harris et al. “The Gene Ontology (GO) database and informatics resource”. en. In: *Nucleic Acids Res.* 32.Database issue (Jan. 2004), pp. D258–61. ISSN: 0305-1048, 1362-4962. DOI: [10.1093/nar/gkh036](https://doi.org/10.1093/nar/gkh036) (cit. on p. 1).
- [14] I. Horrocks et al. *SWRL: A Semantic Web Rule Language Combining OWL and RuleML*. <https://www.w3.org/Submission/SWRL/>. May 2004 (cit. on p. 7).
- [15] S. Kamara, C. Papamanthou, and T. Roeder. “Dynamic searchable symmetric encryption”. In: *Proceedings of the 2012 ACM conference on Computer and communications security*. CCS ’12. Raleigh, North Carolina, USA: Association for Computing Machinery, Oct. 2012, pp. 965–976. ISBN: 9781450316514. DOI: [10.1145/2382196.2382298](https://doi.org/10.1145/2382196.2382298) (cit. on p. 3).
- [16] J. P. McCrae. *The linked open data cloud*. en. <https://lod-cloud.net/>. Accessed: 2022-2-8 (cit. on p. 1).
- [17] G. A. Miller. “WordNet: a lexical database for English”. In: *Commun. ACM* 38.11 (Nov. 1995), pp. 39–41. ISSN: 0001-0782. DOI: [10.1145/219717.219748](https://doi.org/10.1145/219717.219748) (cit. on p. 1).
- [18] MongoDB, Inc. *MongoDB Manual: Encryption at rest*. en. <https://docs.mongodb.com/manual/core/security-encryption-at-rest/>. Accessed: 2022-2-9 (cit. on p. 2).
- [19] M. A. Musen and Protégé Team. “The Protégé Project: A Look Back and a Look Forward”. en. In: *AI Matters* 1.4 (June 2015), pp. 4–12. ISSN: 2372-3483. DOI: [10.1145/2757001.2757003](https://doi.org/10.1145/2757001.2757003) (cit. on p. 2).
- [20] R. Neches et al. “Enabling technology for knowledge sharing”. In: *AI Magazine* 12.3 (1991). ISSN: 0738-4602 (cit. on p. 1).
- [21] N. Noy et al. “Industry-scale knowledge graphs: Lessons and challenges”. In: *Commun. ACM* 62.8 (2019). ISSN: 0001-0782, 1557-7317. DOI: [10.1145/3331166](https://doi.org/10.1145/3331166) (cit. on p. 1).
- [22] Oracle Corporation and/or its affiliates. *MySQL :: MySQL 8.0 reference manual :: 15.13 InnoDB data-at-rest encryption*. en. <https://dev.mysql.com/doc/refman/8.0/en/innodb-data-encryption.html>. Accessed: 2022-2-9 (cit. on p. 2).
- [23] Oracle Corporation and/or its affiliates. *MySQL enterprise encryption*. en. <https://www.mysql.com/products/enterprise/encryption.html>. Accessed: 2022-2-9 (cit. on p. 2).
- [24] W. Pearson. *Cryptography and Network Security: Principles and Practice, Global Edition*. 7th ed. Pearson Education, 2017. ISBN: 9781292158587 (cit. on p. 3).
- [25] E. Prud’hommeaux and A. Seaborn. *SPARQL Query Language for RDF*. <https://www.w3.org/TR/rdf-sparql-query/>. Jan. 2008 (cit. on p. 3).
- [26] Security Compass. *The 2021 State of Cloud Adoption*. en. <https://resources.securitycompass.com/reports/2021-state-of-cloud-adoption>. Accessed: 2022-2-9. Sept. 2021 (cit. on p. 2).
- [27] N. Sioutos et al. “NCI Thesaurus: a semantic model integrating cancer-related clinical and molecular information”. en. In: *J. Biomed. Inform.* 40.1 (Feb. 2007), pp. 30–43. ISSN: 1532-0464, 1532-0480. DOI: [10.1016/j.jbi.2006.02.013](https://doi.org/10.1016/j.jbi.2006.02.013) (cit. on p. 1).

- [28] D. X. Song, D. Wagner, and A. Perrig. “Practical techniques for searches on encrypted data”. In: *Proceeding 2000 IEEE Symposium on Security and Privacy. S P 2000*. May 2000, pp. 44–55. doi: [10.1109/SECPRI.2000.848445](https://doi.org/10.1109/SECPRI.2000.848445) (cit. on p. 3).
- [29] R. Studer, V. R. Benjamins, and D. Fensel. “Knowledge engineering: Principles and methods”. In: *Data Knowl. Eng.* 25.1 (Mar. 1998), pp. 161–197. issn: 0169-023X. doi: [10.1016/S0169-023X\(97\)00056-6](https://doi.org/10.1016/S0169-023X(97)00056-6) (cit. on p. 10).
- [30] The W3C SPARQL Working Group. *SPARQL 1.1 Overview*. <https://www.w3.org/TR/sparql11-overview/>. Mar. 2013 (cit. on p. 3).
- [31] T. Tudorache, J. Vendetti, and N. F. Noy. “Web-Protégé: A lightweight OWL ontology editor for the web”. In: *CEUR Workshop Proceedings*. Vol. 432. 2009 (cit. on p. 2).
- [32] T. Tudorache et al. “Supporting collaborative ontology development in Protégé”. In: *Lect. Notes Comput. Sci.* 5318 LNCS (2008), pp. 17–32. issn: 0302-9743, 1611-3349. doi: [10.1007/978-3-540-88564-1_2](https://doi.org/10.1007/978-3-540-88564-1_2) (cit. on p. 2).
- [33] W3C. *Extensible markup language (XML) 1.0 (fifth edition)*. en. <https://www.w3.org/TR/xml/>. Accessed: 2022-2-11 (cit. on p. 7).
- [34] W3C. *OWL 2 web ontology language direct semantics (second edition)*. en. <https://www.w3.org/TR/2012/REC-owl2-direct-semantics-20121211/>. Accessed: 2022-2-14 (cit. on pp. 10, 11).
- [35] W3C. *OWL 2 web ontology language document overview (second edition)*. en. <https://www.w3.org/TR/owl2-overview/>. Accessed: 2022-2-11 (cit. on pp. 7, 10).
- [36] W3C. *OWL 2 web ontology language Manchester syntax (second edition)*. en. <https://www.w3.org/TR/2012/NOTE-owl2-manchester-syntax-20121211/>. Accessed: 2022-2-14 (cit. on p. 11).
- [37] W3C. *OWL 2 web ontology language RDF-based semantics (second edition)*. en. <https://www.w3.org/TR/2012/REC-owl2-rdf-based-semantics-20121211/>. Accessed: 2022-2-14 (cit. on p. 10).
- [38] W3C. *OWL 2 web ontology language structural specification and functional-style syntax (second edition)*. en. <https://www.w3.org/TR/2012/REC-owl2-syntax-20121211/>. Accessed: 2022-2-14 (cit. on p. 11).
- [39] W3C. *OWL 2 web ontology language XML serialization (second edition)*. en. <https://www.w3.org/TR/2012/REC-owl2-xml-serialization-20121211/>. Accessed: 2022-2-14 (cit. on p. 11).
- [40] W3C. *RDF 1.1 concepts and abstract syntax*. en. <https://www.w3.org/TR/rdf11-concepts/>. Accessed: 2022-2-11 (cit. on pp. 7, 8).
- [41] W3C. *RDF 1.1 TriG*. en. <https://www.w3.org/TR/trig/>. Accessed: 2022-2-13 (cit. on p. 8).
- [42] W3C. *RDF 1.1 turtle*. en. <https://www.w3.org/TR/turtle/>. Accessed: 2022-2-13 (cit. on p. 8).
- [43] W3C. *RDF 1.1 XML syntax*. en. <https://www.w3.org/TR/rdf-syntax-grammar/>. Accessed: 2022-2-13 (cit. on p. 9).
- [44] W3C. *RDF Schema 1.1*. en. <https://www.w3.org/TR/rdf-schema/>. Accessed: 2022-2-11 (cit. on pp. 7, 9).

- [45] W3C. *RDfa core 1.1 - third edition*. en. <https://www.w3.org/TR/rdfa-core/>. Accessed: 2022-2-13 (cit. on p. 9).
- [46] W3C. *Semantic Web Case Studies and Use Cases*. en. <https://www.w3.org/2001/sw/sweo/public/UseCases/>. Accessed: 2022-2-8 (cit. on p. 1).
- [47] Wikipedia contributors. *Semantic web stack*. https://en.wikipedia.org/wiki/File:Semantic_web_stack.svg. Accessed: NA-NA-NA (cit. on p. 7).